Data Transformation

1. Property Price Category

Dim_Property mein:

Add Column → Conditional Column

Column name:

Price_Category

Rules:

```
If List_Price < 5000000
   Affordable
Else if List_Price < 10000000
   Mid Range
Else if List_Price < 20000000
   Premium
Else
   Luxury
```

---

2. Area Category

Column:

Area_Category

```
Area < 800
   Small
Area < 1500
   Medium
Area < 2500
   Large
Otherwise
   Very Large
```

---

3. Property Segment

Create another conditional column:

Property_Segment

Logic:

```
If Property_Type = Apartment
   Residential
If Property_Type = Villa
   Residential
If Property_Type = Row House
   Residential
If Property_Type = Plot
   Residential
Otherwise
   Commercial
```

---

4. Sales — Discount Category

Fact_Sales

Create:

Discount_Category

Logic:

```
Discount_Pct = 0
   No Discount
Discount_Pct <= 5
   Low Discount
Discount_Pct <= 8
   Medium Discount
Otherwise
   High Discount
```

---

5. Agent Experience Category

Dim_Agent

Create:

Experience_Category

Logic:

```
Experience_Years <= 2
  Fresher
Experience_Years <= 5
  Junior
Experience_Years <= 8
  Experienced
Otherwise
  Senior
```

---

6. Customer Age Group

Dim_Customer

Create:

Age_Group

Logic:

```
Age < 30
  18-29
Age < 40
  30-39
Age < 50
  40-49
Otherwise
  50+
```

---

7. Marketing — Cost Per Lead

Fact_Marketing

Add Column → Custom Column

Formula:

```
[Marketing_Spend] / [Leads_Generated]
```

Column:

Cost_Per_Lead

---

8. Marketing — Conversion Rate

Custom Column:

```
[Conversions] / [Leads_Generated]
```

Column:

Conversion_Rate

Merge Queries

1. Property + Location

Merge **Dim_Property + Dim_Location** using Location_ID → **Left Outer** → Create New Query → **Property_Analysis**

2. Leads + Campaign

Merge **Fact_Leads + Dim_Campaign** using Campaign_ID → **Left Outer** → Create New Query → **Lead_Campaign_Analysis**

3. Leads + Customer

Merge **Fact_Leads + Dim_Customer** using Customer_ID → **Left Outer** → Create New Query → **Lead_Customer_Analysis**

4. Sales + Property

Merge **Fact_Sales + Dim_Property** using Property_ID → **Left Outer** → Create New Query → **Sales_Property_Analysis**

Star Schema & Relationships

Open **Model View**.

Step 1 — Fact & Dimension Tables

Dimensions:
Dim_Date, Dim_Property, Dim_Customer, Dim_Agent, Dim_Location, Dim_Campaign

Facts:
Fact_Sales, Fact_Leads, Fact_Visits, Fact_Payments, Fact_Marketing

Rule: Dimension = Who / What / Where / When | Fact = Business Event / Transaction

---

Step 2 — Fact_Sales Relationships

Create these relationships:

- Dim_Date[Date] → Fact_Sales[Sale_Date] → *1:* | *Single* | *Active**
- Dim_Customer[Customer_ID] → Fact_Sales[Customer_ID] → *1:* | *Single* | *Active**
- Dim_Property[Property_ID] → Fact_Sales[Property_ID] → *1:* | *Single* | *Active**
- Dim_Agent[Agent_ID] → Fact_Sales[Agent_ID] → *1:* | *Single* | *Active**
- Dim_Location[Location_ID] → Fact_Sales[Location_ID] → *1:* | *Single* | *Active**

Step 3 — Fact_Leads Relationships

- Dim_Date[Date] → Fact_Leads[Lead_Date] → *1:* | *Single* | *Active**
- Dim_Customer[Customer_ID] → Fact_Leads[Customer_ID] → *1:* | *Single* | *Active**
- Dim_Property[Property_ID] → Fact_Leads[Property_ID] → *1:* | *Single* | *Active**
- Dim_Agent[Agent_ID] → Fact_Leads[Agent_ID] → *1:* | *Single* | *Active**
- Dim_Campaign[Campaign_ID] → Fact_Leads[Campaign_ID] → *1:* | *Single* | *Active**

Step 4 — Fact_Visits Relationships

- Dim_Date[Date] → Fact_Visits[Visit_Date] → *1:* | *Single* | *Active**
- Dim_Customer[Customer_ID] → Fact_Visits[Customer_ID] → *1:* | *Single* | *Active**
- Dim_Property[Property_ID] → Fact_Visits[Property_ID] → *1:* | *Single* | *Active**
- Dim_Agent[Agent_ID] → Fact_Visits[Agent_ID] → *1:* | *Single* | *Active**

Step 5 — Fact_Payments

Create:

Fact_Sales[Sale_ID] → Fact_Payments[Sale_ID]

*1: | Single | Active**

Reason: **One Sale → Multiple Payments**

Step 6 — Fact_Marketing

Create:

Dim_Campaign[Campaign_ID] → Fact_Marketing[Campaign_ID]

*1: | Single | Active**

For now, don't create active Date relationships for Start_Date and End_Date.